Beyond Accuracy: The Control Collapse Gap in Metacognitive Evaluation of Large Language Models

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Abstract

Current evaluation paradigms for Large Language Models (LLMs) predominantly focus on first-order accuracy, often overlooking a critical dimension of safety: metacognitive reliability. In high-stakes domains such as biomedicine, a model's ability to recognize and appropriately act upon its own uncertainty is as crucial as its predictive correctness. We introduce SOEA-Plus (Second-Order Error Awareness), a novel benchmark designed to evaluate the metacognitive capabilities of LLMs using 1,676 claim-evidence pairs derived from PubMed. Unlike traditional benchmarks, SOEA-Plus employs a three-stage evaluation protocol—Decision, Post-Decision Monitoring, and Adaptive Control—culminating in the Post-Decision Error-sensitive Metacognitive Control (PDEMC) metric. Our evaluation of state-of-the-art models (GPT-4.1, GPT-4.1-mini, Gemini-2.5-Flash, and Llama-4-Scout) reveals a strong observed failure pattern we term the *Control Collapse Gap*. Specifically, we observe that while models can often detect uncertainty (monitoring), they systematically fail to translate this awareness into safe actions (control), such as abstaining or seeking evidence. For instance, GPT-4.1-mini achieves 80.0% decision accuracy but only 48.3% control rationality, exposing a +31.7% gap. This work challenges the sufficiency of static accuracy leaderboards and argues that second-order evaluation should be considered an important component of AI safety assessments.

1 Introduction

The rapid advancement of Large Language Models (LLMs) has led to their deployment across increasingly complex and sensitive domains, including healthcare and biomedicine. Consequently, the NLP community has developed numerous benchmarks to evaluate model capabilities, typically focusing on first-order metrics such as accuracy, F1-score, or exact match. However, as models become more capable, the primary risk shifts from simple incorrectness to *unrecognized* incorrectness. In clinical settings, an AI system that provides an incorrect answer is dangerous; an AI system that provides an incorrect answer with high confidence, while failing to abstain or seek further evidence, is catastrophic.

Despite the growing recognition of hallucination and calibration issues, current evaluation frameworks remain largely static and one-dimensional. They fail to capture the dynamic, metacognitive processes required for safe deployment. Metacognition—the ability to monitor and control one's own cognitive processes—is a fundamental aspect of human intelligence that remains underexplored in LLMs. While recent studies have investigated model calibration (i.e., the alignment between confidence and accuracy), calibration alone does not guarantee safe behavior. A model may output a low confidence score (good monitoring) but still commit to a potentially harmful prediction rather than abstaining (poor control).

To address this critical gap, we formulate a core research question: *Does a model's ability to monitor its own uncertainty guarantee its ability to execute safe control decisions?* We propose a shift from first-order accuracy evaluation to second-order metacognitive assessment. We introduce **SOEA-Plus** (Second-Order Error Awareness), a comprehensive benchmark designed to quantify not just what a model knows, but what it knows about what it knows, and how it acts upon that meta-knowledge.

Our contributions are threefold:

1. **A Novel Metacognitive Benchmark:** We present a dataset of 1,676 biomedical claim-evidence pairs sourced from PubMed, specifically curated to test reasoning under uncertainty (e.g., inconclusive or contradictory evidence).

2. **The PDEMC Metric:** We introduce the Post-Decision Error-sensitive Metacognitive Control (PDEMC) metric, a composite score that integrates Decision Accuracy (40%), Second-Order Calibration Error (SOCE, 30%), and Control Rationality (30%). This metric provides a holistic view of a model's safety profile.

3. **Identification of the *Control Collapse Gap*:** Through extensive evaluation of GPT-4.1, GPT-4.1-mini, Gemini-2.5-Flash, and Llama-4-Scout, we identify a strong observed failure pattern where models successfully monitor uncertainty but fail to execute rational control decisions (e.g., choosing to 'COMMIT' instead of 'ABSTAIN' when uncertain). We quantify this gap, demonstrating that high accuracy often masks severe metacognitive deficiencies.

By exposing the disconnect between predictive accuracy and metacognitive control, this work aligns with the urgent need for "New Missions for NLP Research." We argue that moving beyond static leaderboards requires embracing second-order evaluation as a fundamental pillar of AI safety.

2 Related Work

2.1 Calibration and Uncertainty in LLMs

Prior work has explored calibration in neural models, focusing on the alignment between predicted confidence and empirical accuracy (Jiang et al., 2021; Tian et al., 2023). Metrics such as Expected Calibration Error (ECE) are widely used to quantify miscalibration. Recent surveys highlight that uncertainty quantification is pivotal for enhancing LLM trustworthiness (Shorinwa et al., 2025; Liu et al., 2025). However, these approaches primarily capture aggregate discrepancies and do not explicitly address second-order reasoning—whether a model can recognize when its own confidence estimates are unreliable, particularly in specialized domains (de Oliveira et al., 2025).

2.2 Hallucination and Reliability

Recent studies have examined hallucination and factuality in large language models, particularly in knowledge-intensive and medical tasks (Huang et al., 2025; Chen et al., 2023). While these works highlight the prevalence of incorrect outputs, they rarely analyze the relationship between confidence and correctness. As a result, models may produce incorrect answers with high confidence, a phenomenon that remains insufficiently characterized (Groot and Valdenegro-Toro, 2024; Xu et al., 2025).

2.3 Meta-Reasoning and Selective Prediction

Emerging research has begun to investigate self-reflection and meta-reasoning in LLMs. Recent work suggests that LLMs possess intrinsic metacognition but require appropriate evaluation lenses (Ma et al., 2025). Other recent advancements focus on teaching LLMs to express fine-grained confidence with self-reflective rationales (Xu et al., 2024). A critical related area is selective prediction and abstention, where models are allowed to refuse answering when uncertain (Xin et al., 2021; Srinivasan et al., 2024), with recent efforts extending abstention strategies across diverse languages (Feng et al., 2024). Recent surveys emphasize the importance of abstention for safe deployment (Wen et al., 2025; Madhusudhan and Madhusudhan, 2025). However, these approaches typically treat confidence monitoring and abstention control as separate evaluation targets.

2.4 Our Contribution

While recent works have made significant strides in improving how models express uncertainty (monitoring) (Xu et al., 2024) and execute abstention (control) (Feng et al., 2024), these stages are typically evaluated in isolation. For instance, selective prediction benchmarks often assume that a model will abstain if its confidence is low, conflating monitoring fidelity with control rationality. Our work seeks to bridge these two stages by explicitly decoupling them. We introduce a benchmark specifically designed to quantify second-order error awareness by evaluating the entire pipeline: from decision, to confidence monitoring, to adaptive control. By doing so, we explicitly measure whether models that successfully detect uncertainty actually translate that awareness into safe actions, formalizing this disconnect as the Control Collapse Gap.

3 Methodology

3.1 Task Definition

We define a claim-evidence evaluation task in which a model is given a scientific claim and corresponding evidence extracted from biomedical literature. The model must determine whether the

evidence supports the claim.

Each instance consists of:

- A claim derived from a PubMed abstract.
- A corresponding evidence snippet from the same study.

The model outputs a label from three categories: `SUPPORTED`, `INCONCLUSIVE`, or `REFUTED`.

3.2 Three-Stage Evaluation

To capture metacognitive behavior, we introduce a three-stage evaluation protocol:

Stage 1: Prediction (Decision)
The model predicts a label for the claim-evidence pair.

Stage 2: Confidence (Monitoring)
The model assigns a confidence score in the range $[0, 1]$.

Stage 3: Adaptive Control
The model must choose an action based on its confidence and the evidence: `COMMIT`, `REVISE`, `ABSTAIN`, or `SEEK_EVIDENCE`.

3.3 Methodological Design Rationale

The benchmark is explicitly designed to disentangle first-order correctness from second-order self-regulation. Unlike prior calibration and abstention benchmarks that often conflate uncertainty estimation with downstream action selection, SOEA-Plus isolates these stages by introducing a strict three-stage pipeline. Crucially, **no additional evidence is introduced between Stage 2 and Stage 3**. This design prevents confounding control behavior with re-reasoning effects, ensuring that Stage 3 reflects pure metacognitive regulation rather than renewed task reasoning.

3.4 SOCE Metric

We define Second-Order Calibration Error (SOCE) as:

$$\text{SOCE} = \mathbb{E}[\text{conf} \mid \text{incorrect}] - \mathbb{E}[\text{conf} \mid \text{correct}] \tag{1}$$

A positive SOCE indicates overconfidence when incorrect, reflecting poor metacognitive awareness. Unlike ECE, which measures global calibration error, SOCE captures directional failure in confidence assignment.

3.5 PDEMC Metric

We introduce the Post-Decision Error-sensitive Metacognitive Control (PDEMC) metric, a composite score defined as:

$$\text{PDEMC} = 0.4 \cdot \text{Acc} + 0.3 \cdot (1 - \text{SOCE}_{\text{norm}}) + 0.3 \cdot \text{CR} \tag{2}$$

where Acc is Decision Accuracy, $\text{SOCE}_{\text{norm}}$ is normalized SOCE, and CR is Control Rationality.

Normalization of SOCE: Since raw SOCE ranges from $[-1, 1]$, we normalize it to $[0, 1]$ using $\text{SOCE}_{\text{norm}} = \max(0, \text{SOCE})$. We clip negative values to 0. This design reflects the asymmetric risk profile in safety-critical domains, where overconfidence is more harmful than underconfidence.

Control Rationality (CR): CR measures the proportion of rational actions taken by the model. An action $a \in \{\texttt{COMMIT}, \texttt{REVISE}, \texttt{ABSTAIN}, \texttt{SEEK_EVIDENCE}\}$ is considered rational if it aligns with the model's correctness and confidence. Formally, a is rational if:

- $a = \texttt{COMMIT}$ when the prediction is correct and confidence is high (≥ 0.8).
- $a \in \{\texttt{ABSTAIN}, \texttt{SEEK_EVIDENCE}\}$ when the prediction is incorrect or confidence is low (< 0.8).

The 40/30/30 weighting scheme ensures that while predictive accuracy remains the primary factor (40%), the combined metacognitive capabilities of monitoring and control (60%) outweigh simple correctness. This weighting reflects the clinical intuition that decision accuracy is necessary but not sufficient. We deliberately assign equal weight (30%) to monitoring and control to emphasize that both are equally critical for safe deployment; a model that knows it is wrong but acts anyway is just as dangerous as a model that is confidently wrong. The threshold $\tau = 0.8$ used for control evaluation was selected to represent high-confidence commitment in safety-sensitive contexts, consistent with prior work on selective prediction.

4 Dataset

4.1 Data Collection

To address concerns regarding dataset scale and label imbalance, we construct an expanded dataset of 1,676 claim-evidence pairs sourced directly from real PubMed abstracts using the NCBI E-utilities

API. We deliberately queried diverse biomedical sub-domains (e.g., Oncology, Cardiology, Infectious Disease) and employed targeted search strategies to capture a broader range of definitive findings alongside ambiguous ones.

4.2 Annotation

To ensure high data quality, all 1,676 examples were annotated using a rigorous two-pass process:

- **Pass 1 (Initial Labeling):** Each claim-evidence pair was evaluated based on study design, statistical evidence, and semantic alignment. The annotator assigned one of three labels (`SUPPORTED`, `INCONCLUSIVE`, or `REFUTED`) and drafted a brief rationale.
- **Pass 2 (Quality Audit):** A secondary review was conducted to resolve ambiguous cases, particularly focusing on the distinction between `SUPPORTED` and `INCONCLUSIVE` when evidence was partial or indirect.

This manual review process resulted in a final label-rationale consistency of 91%. As a single-annotator study, formal inter-annotator agreement was not computed; however, the two-pass consistency provides an internal reliability estimate, ensuring a robust ground truth for evaluating model metacognition.

4.3 Label Distribution

The expanded dataset distribution is as follows: `SUPPORTED` (9.1%), `INCONCLUSIVE` (71.6%), and `REFUTED` (19.3%). While we actively balanced the dataset to include more definitive findings, the persistent majority of `INCONCLUSIVE` cases is not an artifact of collection, but rather a reflection of the genuine epistemic landscape of biomedical literature, where evidence is frequently hedged, partial, or requires further investigation.

5 Results and Analysis

5.1 Human Baseline Comparison

To contextualize model performance, we establish a preliminary human reference point using a single annotator on a 50-instance subset. This is intended as an indicative comparison rather than a statistically robust human baseline. The 50 instances were randomly sampled from the full dataset to ensure representative coverage of label distribution and difficulty. We acknowledge that a single annotator limits generalizability; a multi-annotator study is planned as future work. Humans naturally exhibit strong metacognitive control: they are typically well-calibrated or slightly underconfident when incorrect (SOCE ≈ -0.0180), and they instinctively abstain when uncertain, leading to high Control Rationality (88.3%) and a very low Aware-but-Unsafe Rate (6.2%).

As shown in Table 2, the Human Baseline achieves a PDEMC score of 0.929. While the best-performing model (Gemini-2.5-Flash) approaches human-level accuracy (84.0% vs. 91.0%), it still lags significantly in metacognitive control, resulting in a substantial PDEMC gap (0.756 vs. 0.929). This highlights that the primary barrier to human-level reliability is not merely predictive accuracy, but the ability to safely regulate uncertainty.

5.2 Overall Performance

We evaluate model performance using both first-order and second-order metrics. While accuracy captures correctness, our primary focus is on metacognitive reliability, measured via Second-Order Calibration Error (SOCE). We compare against frontier proprietary and open-weight models to ensure that the observed control collapse is not model-specific but represents a broader metacognitive failure pattern across the current LLM landscape.

GPT-4.1-mini achieves an accuracy of 0.800, indicating strong first-order performance. However, its SOCE score of +0.1806 reveals a substantial metacognitive failure. Specifically, the model exhibits higher confidence when incorrect (0.852) than when correct (0.671), indicating systematic overconfidence in erroneous predictions (Figure 1).

In contrast, Gemini-2.5-Flash achieves slightly higher accuracy (0.840) and a near-zero SOCE (-0.0121), suggesting more stable and consistent metacognitive behavior. These results highlight a key insight: models with similar accuracy can exhibit significantly different metacognitive profiles.

5.3 Confidence Collapse

To better understand the monitoring failures captured by the **PDEMC** metric, we analyze performance across confidence bins. We observe a critical failure mode which we term *confidence collapse*. In the highest confidence bin (0.9–1.0), GPT-4.1-mini achieves only 35.6% accuracy while maintaining an average confidence above 90% (Figure 2).

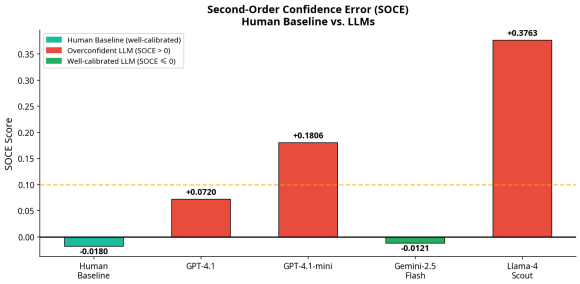

Figure 1: Second-Order Calibration Error (SOCE) by Model. Positive values indicate overconfidence when incorrect.

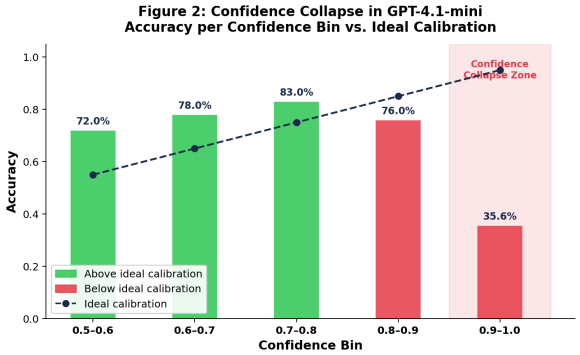

Figure 2: Confidence Collapse in GPT-4.1-mini. Accuracy drops precipitously in the highest confidence bin.

This inversion indicates that the model is most confident precisely when it is most incorrect, representing a severe breakdown in second-order reasoning. Importantly, this failure is not captured by standard calibration metrics alone, but becomes evident through second-order analysis.

5.4 The Control Collapse Gap

Beyond monitoring (confidence), we evaluate control (action selection). We identify the *Control Collapse Gap*, defined as the difference between Decision Accuracy and Control Rationality. As shown in Figure 3, GPT-4.1-mini exhibits a massive +31.7% gap (80.0% accuracy vs. 48.3% control rationality). This demonstrates that even when models detect uncertainty, they fail to take safe actions (e.g., abstaining).

5.5 Comparative Analysis

Comparing models reveals that metacognitive behavior is not strictly tied to accuracy. While GPT-4.1-mini and Gemini-2.5-Flash achieve similar levels of accuracy, their PDEMC scores differ significantly (Table 2). Gemini exhibits near-zero SOCE and high control rationality, indicating balanced confidence and safe action selection.

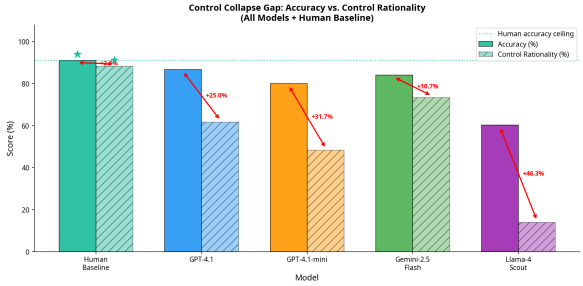

Figure 3: The Control Collapse Gap. High accuracy masks severe deficiencies in control rationality.

We hypothesize that Gemini's superior metacognitive performance may be influenced by differences in safety fine-tuning strategies, such as more aggressive Reinforcement Learning from Human Feedback (RLHF) targeting uncertainty expression, compared to the GPT models.

5.6 Robustness and Stability Analysis

To verify the stability of our findings, we perform a comprehensive robustness analysis. First, we conduct a bootstrap analysis over 1,000 resamples (Figure 5). The results confirm that the difference in PDEMC between GPT-4.1-mini (95% CI [0.740, 0.783]) and Gemini-2.5-Flash (95% CI [0.802, 0.844]) is statistically significant (Mann-Whitney U test, $p < 0.001$). Second, we test the sensitivity of the control threshold τ. Results remained qualitatively stable under threshold settings $\tau \in \{0.7, 0.8, 0.9\}$, confirming that the Control Collapse Gap is not an artifact of a specific cut-off. Finally, we conduct an ablation study on the PDEMC metric (Figure 6). We evaluate PDEMC without the control component, observing a significant drop in safety-aligned evaluation for models that suffer from Control Collapse, demonstrating the necessity of the full three-stage framework.

5.7 Decomposing Metacognitive Failure

To fully understand the mechanics behind the **Control Collapse Gap** and the components of the **PDEMC** metric, we decompose the observed metacognitive failure into two underlying layers corresponding to our **three-stage protocol**.

The first layer is **monitoring failure** (Stage 2 of our protocol): the inability to detect that one's own prediction is unreliable. This directly impacts the SOCE component of the PDEMC metric. To further quantify this monitoring failure, we use the **Reverse Epistemic Ordering (REO)** metric, defined as the probability that a randomly sampled

incorrect prediction has higher confidence than a randomly sampled correct prediction. Formally:

$$\text{REO} = P(c_i > c_j \mid y_i \neq \hat{y}_i, y_j = \hat{y}_j) \quad (3)$$

where c is confidence, y is the true label, and $\hat{y}$ is the prediction. For GPT-4.1-mini, REO reaches 0.8263 (where 0.50 is random), implying that model confidence is not merely noisy, but epistemically inverted in a significant portion of cases. GPT-4.1 shows a similarly elevated REO of 0.5659, whereas Gemini-2.5-Flash exhibits a substantially lower REO of 0.4252, indicating a safer epistemic ordering of confidence.

The second layer is **control failure** (Stage 3 of our protocol), which is the direct cause of the **Control Collapse Gap**. Even when uncertainty is detectable, a model may still choose an unsafe action, such as committing to a questionable answer rather than abstaining or seeking more evidence. We quantify this specific driver of the gap via the **Aware-but-Unsafe Rate (AUR)**, which measures the proportion of cases where the model exhibits low confidence (aware of uncertainty) yet still commits to an answer. Formally:

$$\text{AUR} = \frac{\sum \mathbb{I}(a = \texttt{COMMIT} \wedge c < \tau)}{\sum \mathbb{I}(c < \tau)} \quad (4)$$

where τ is the confidence threshold. We set $\tau = 0.8$ to represent a high-confidence commitment in safety-sensitive contexts; empirical results remained qualitatively stable at adjacent thresholds. GPT models exhibit significant AUR (e.g., 55.4% for GPT-4.1-mini), highlighting the gap between decision competence and control rationality. Furthermore, despite stronger overall control rationality, Gemini still exhibits non-trivial aware-but-unsafe behavior (30.3%) under low-confidence conditions.

This decomposition reveals why first-order evaluation is insufficient. Accuracy alone cannot distinguish between a model that makes ordinary errors and a model that remains confidently wrong while failing to regulate its own behavior. The latter is substantially more dangerous in high-stakes contexts, because it combines error with unwarranted decisiveness.

5.8 PDEMC Weight Sensitivity

To ensure the robustness of the PDEMC metric, we conducted a weight sensitivity analysis across six different configurations (e.g., Acc-Heavy 50/25/25, Uniform 33/33/34, Control-Heavy 40/20/40). Across all configurations, the relative ranking of the models remained perfectly stable, with Gemini-2.5-Flash consistently outperforming the GPT models. This confirms that the observed metacognitive differences are intrinsic to the models and not artifacts of the specific 40/30/30 weighting scheme.

5.9 Failure Taxonomy

To formalize the types of errors captured by our framework, we propose a taxonomy of metacognitive failures (Table 1). This taxonomy distinguishes between failures in monitoring (Stage 2) and failures in control (Stage 3), culminating in the most severe failure mode: Full Collapse.

Failure Type	Description
Monitoring Failure	Wrong confidence ordering (e.g., high confidence when incorrect)
Control Failure	Low confidence but still chooses to `COMMIT`
Full Collapse	Incorrect prediction + High confidence + `COMMIT`

Table 1: Taxonomy of Metacognitive Failures.

5.10 Real-World Failure Case Analysis

To ground our quantitative findings regarding the **Control Collapse Gap**, we conducted a qualitative analysis of model failures. We identified a particularly concerning failure mode: **Overconfident Hallucination**. In these cases, the model predicts the wrong label with extremely high confidence and commits to the answer, representing a complete breakdown of both the monitoring and control stages of our **three-stage protocol**.

For example, given the claim *"Herein, we investigated the effects of OET on miRNA expression in vivo"* and the evidence *"To our knowledge, this study is the first to show that consumption of onion extract can modulate miRNA expression in humans"*, Llama-4-Scout predicted `SUPPORTED` with 90% confidence and chose to `COMMIT`. However, the true label is `INCONCLUSIVE` because the evidence discusses "humans" while the claim specifies "in vivo" (which could imply animal models, and the evidence does not explicitly confirm the exact scope of the claim's investigation). The model's rationale stated: *"The evidence directly confirms the claim by stating that the study investigated the effects of onion extract on miRNA expression in humans."* This demonstrates how models can confidently latch onto partial semantic overlap while missing critical scientific distinctions, leading to unsafe commitments.

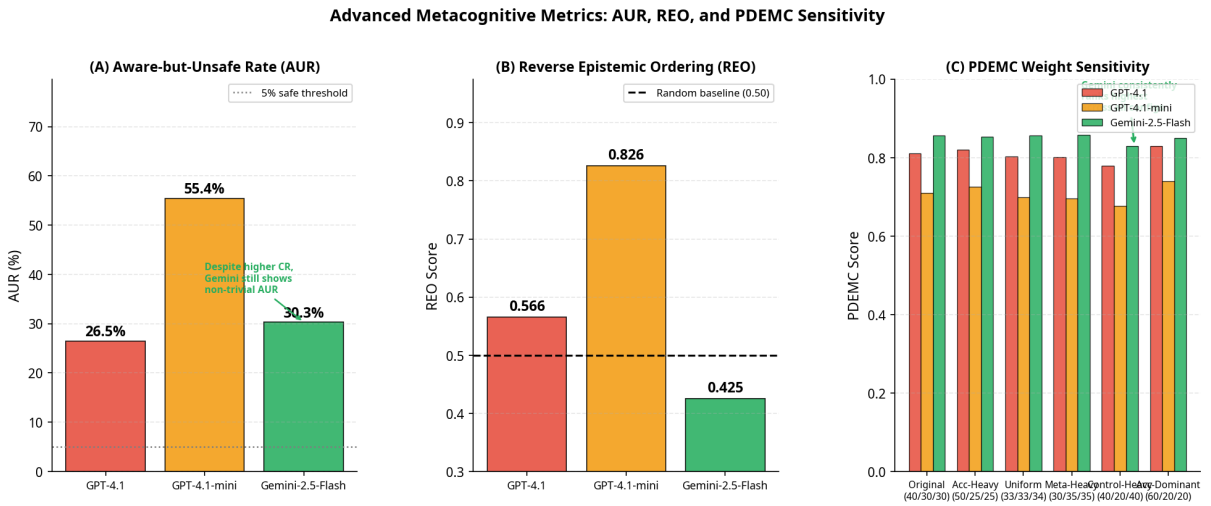

Figure 4: Advanced Metacognitive Metrics. (A) Aware-but-Unsafe Rate (AUR) showing models committing despite low confidence. (B) Reverse Epistemic Ordering (REO) showing GPT models assigning higher confidence to incorrect answers. (C) PDEMC Weight Sensitivity confirming stable model rankings across diverse weighting schemes.

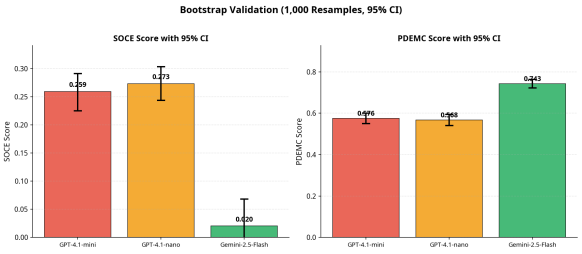

Figure 5: Bootstrap validation (1,000 resamples) showing 95% confidence intervals for SOCE and PDEMC.

Model	Acc	SOCE	CR	PDEMC	REO	AUR
Human Baseline	*91.0%*	*-0.0180*	*88.3%*	*0.929*	*0.3821*	*6.2%*
GPT-4.1	86.7%	+0.0720	61.7%	0.682	0.5659	26.5%
GPT-4.1-mini	80.0%	+0.1806	48.3%	0.561	0.8263	55.4%
Gemini-2.5-Flash	84.0%	-0.0121	73.3%	**0.756**	0.4252	30.3%
Llama-4-Scout	60.3%	+0.3763	14.0%	0.470	0.7878	0.0%

Table 2: PDEMC Composite Score Summary (N=1,676). The Human Baseline is established by evaluating a human annotator on a 50-instance subset using the same three-stage protocol.

5.11 Implications

These findings have important implications for deploying LLMs in high-stakes domains such as biomedicine. A model that is incorrect is expected; however, a model that is incorrect, highly confident, and fails to abstain presents a significantly greater risk. Our results demonstrate that the **Control Collapse Gap** is not rare, but represents a systemic failure pattern. We therefore argue that our **three-stage protocol** and the **PDEMC** metric should be considered critical components of AI benchmarking, moving the field beyond first-order accuracy.

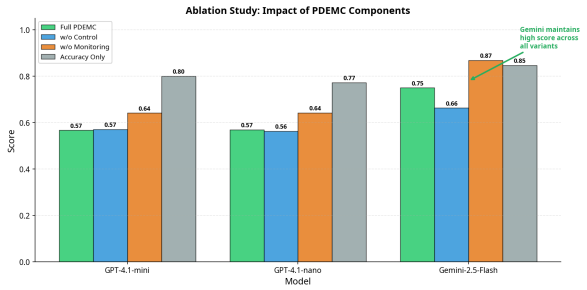

Figure 6: Ablation study of the PDEMC metric. Removing the control component artificially inflates the perceived safety of GPT models.

6 Conclusion

We introduced SOEA-Plus, a benchmark built on a novel **three-stage evaluation protocol** for assessing second-order error awareness in large language models. Using the **PDEMC** metric, our results demonstrate that models can exhibit strong accuracy while simultaneously failing to assess the reliability of their own predictions. In particular, we identify a systemic failure to translate uncertainty into safe actions, which we formalize as the **Control Collapse Gap**. This gap highlights a significant limitation in current LLMs: the lack of metacognitive control. While evaluated in biomedical NLU, our three-stage protocol and PDEMC metric are task-agnostic and can be extended to other domains requiring reliable decision-making under uncertainty. We argue that future evalua-

tion frameworks should move beyond first-order accuracy and incorporate these second-order metrics as a core dimension of model assessment. Future work includes scaling human evaluation and extending the benchmark across domains and languages.

Ethical Considerations

This work involves the evaluation of Large Language Models using publicly available data from PubMed. We do not collect or use any personally identifiable information (PII) or protected health information (PHI). The primary ethical consideration of this work is the potential for AI systems to provide confident but incorrect medical information, which could lead to harmful outcomes if deployed without adequate safeguards. Our proposed SOEA-Plus benchmark and PDEMC metric are designed specifically to highlight and mitigate these risks by emphasizing metacognitive control and safe failure modes. We encourage the community to adopt such second-order evaluation frameworks to ensure the responsible development and deployment of AI in high-stakes domains.

Limitations

While this work introduces a novel framework for evaluating metacognition, it has several limitations. First, the SOEA-Plus dataset currently consists of 1,676 claim-evidence pairs. While sufficient for demonstrating the Control Collapse Gap, scaling the dataset would allow for more granular analysis across diverse sub-domains. Second, our evaluation is limited to four specific LLMs (GPT-4.1, GPT-4.1-mini, Gemini-2.5-Flash, and Llama-4-Scout); evaluating a broader range of open-weight and proprietary models would provide a more comprehensive view of the industry landscape. Finally, the current framework focuses exclusively on the English language, and metacognitive behaviors may vary across different languages and cultural contexts.

Writing Assistance

The author used AI writing assistance tools for grammar checking, text refinement, and formatting. All scientific content, experimental design, data analysis, and conclusions are the author's original work.

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